Centro de Investigación y de Estudios Avanzados del Instituto
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Sensibilidades a radio de carga de neutrinos en experimentos de neutrinos de reactor involucrando CEνNS

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PHYSICS DEPARTMENT

Sensitivities to neutrino charge radius at reactor neutrino experiments involving CEνNS

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agradecimiento aqui.

Resumen

resumen aqui

Abstract

resumen en ingles

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1. Introduction

intro

2. Electroweak theory in the Standard Model of particles

2.1. Introduction

The Standard Model of particles (SM) is the most currently utilized theory of interactions of subatomic particles. It describes all matter particles and forces between them, excluding gravity, as interactions between 12 fermions (6 quarks and 6 leptons) with their corresponding anti-fermions, 4 spin-1 force-carrying gauge bosons and 1 scalar boson as shown in Fig. 2.1. Within this framework exist 3 distinct forces: strong nuclear force, mediated by the gluon (g) boson; electromagnetic (EM) force, mediated by the photon (γ) boson; and weak nuclear force, mediated by the $W^{\pm}$ and Z bosons. This work pertains to the description of the properties of the only family of particles that present no electric or color charge known as neutrinos (ν), wherein the SM exist three known flavors (ν_e, ν_μ and ν_τ) corresponding to the latter half of (electro)weak isospin doublets formed with the known charged leptons (e^-, μ^-, τ^-). This chapter aims to describe neutrinos in the SM, the weak force and its origin as a part of a larger broken symmetry known as "Electroweak Theory" (EW).

2.2. Symmetries and fermion content

2.2.1 Neutrinos

Neutrinos are charge-less, color-less, spin-$\frac{1}{2}$ particles with negligible masses compared to other fundamental particles first proposed in 1930 by Pauli W. [1], and later detected by C. L. Cowan et al. in 1956 [2], to solve known apparent contradictions in conserved quantities of beta decays. Beta decay was known to be as the emission of beta (electron) particles by the nuclei of unstable atoms, wherein the difference in mass of the original and resulting nuclei of the decay should fix the energy at which the emitted particles are detected. Also of note, due to conservation of angular momentum the sum of the spin (S_e) of the electron, the spin of the resulting nucleus (S_{N^*}) and the angular momentum between them (L) should equal the original nucleus' spin (S_N). Despite this, it was know experimentally that the resulting electrons where emitted in a spectrum of energies less than expected and at $L = 0$, making the sum of spins impossible to match. Pauli's solution to the apparent violation of the conservation of these quantities was to hypothesize the existence of a second neutral spin-$\frac{1}{2}$ with very small mass compared to the electron that allows for both the distribution of energy between both emitted particle, thus letting the electron be emitted at different energies depending on the kinematics, and the conservation of

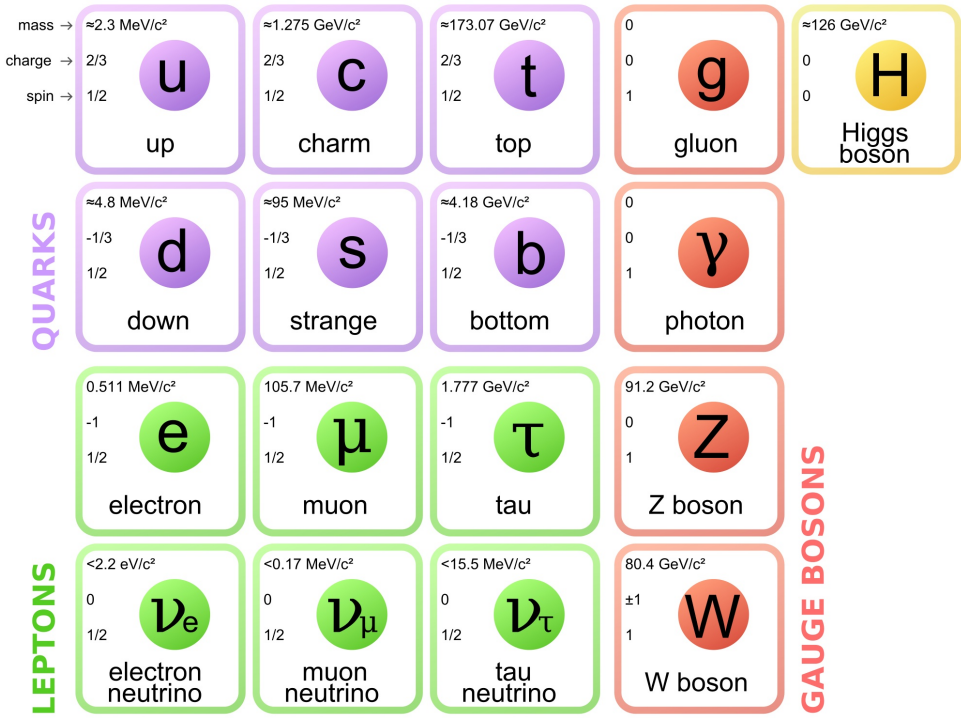

Figure 2.1: *Particles in the SM*

angular momentum.

$N \to N^* + e^-$	$\to$	$N \to N^* + e^- + \nu$
$m_N = m_{N^*} + E_{e^-}$	$\to$	$m_N = m_{N^*} + E_{e^-} + E_\nu$
$S_{z\ N} = S_{z\ N^*} + S_{z\ e^-}$	$\to$	$S_{z\ N} = S_{z\ N^*} + S_{z\ e^-} + S_{z\ \nu}$

Later refinement of the atomic model found that nuclei where made up of positively charged elements named protons (p) and neutrally charges elements known as neutrons[1] (n), leaving the small hypothesized particle to be named neutrino later by Enrico Fermi. Its is well understood now that the beta decay is a process in which a neutron within an unstable nucleus decays into a proton, an electron and a(n) (anti)neutrino.

$$n \to p + e + \nu$$

2.2.2 Weak Nuclear Force

2.3. Massive and Massless Mediators

2.4. Electroweak mixing $\sin^2 \theta_W$

[1]this was the name first proposed for the particle hypothesized by Pauli

3. Radiative Corrections and Neutrino Charge Radius

3.1. Introduction

3.2. Neutrino Charge Radius

3.3. One-loop Corrections to the $\nu\nu\gamma$ vertex

3.4. NCR as a radiative correction of Electroweak Mixing

4. Coherent Elastic Neutrino-Nucleus Scattering

4.1. Introduction

4.2. Cross Section

4.3. Experiments

4.4. Quenching Factors

4.5. Neutrino Sources

4.6. Form Factors

5. SBC-ININ-LAr

5.1. Results

6. Summary and conclusions

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